Self-Learning: Address Decoding Techniques & 8259 Cascading MPCA Module 4

> **Definition: Address Decoding Techniques** define how system address lines are mapped to peripheral chip selects (Absolute vs Partial), while **8259 Cascading** expands interrupt handling capacity from 8 to 64 priority channels.

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1. Absolute vs Partial Address Decoding

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ABSO

PARTIAL ADDRESS DECODING:

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### Comparison Table:

| Feature | Absolute Decoding | Partial Decoding |

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## 2. 8259 Cascaded Mode (Master-Slave Configuration)

To handle more than 8 interrupts, one **Master 8259** is connected to up to **8 Slave 8259s** via the 3-bit Cascade Bus (**CAS_0, CAS_1, CAS_2**).

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Cascading Operation Flow:

1. Slave 8259 receives interrupt on its IR line and signals Master 8259 via Master's IR pin.

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3. Quick Recall Flow

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64 Inte
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